

Math 2650 – Quiz 04 Concept Practice

Choose the section matching the skill you want to practice. Answers are on the back.

A. Position, Velocity, and the Osculating Plane

A particle has acceleration

$$\mathbf{a}(t) = \langle 2 \cos t, 6t + 1, -4 \rangle.$$

At time $t = 0$, its position and velocity are

$$\mathbf{r}(0) = \langle 0, 1, -1 \rangle, \quad \mathbf{v}(0) = \langle 1, 2, 1 \rangle.$$

- (a) Find the position function $\mathbf{r}(t)$ and determine the particle's position at $t = 1$.
- (b) Find an equation of the osculating plane to the particle's path at $t = 0$.

B. Motion with Constant Acceleration

An object moves with constant acceleration

$$\mathbf{a}(t) = \langle 0, g, 0 \rangle.$$

At $t = 0$,

$$\mathbf{r}(0) = \langle 2, 50, -3 \rangle, \quad \mathbf{v}(0) = \langle 6, 0, 8 \rangle.$$

At some later time, the object is at

$$\langle 14, 18, 13 \rangle.$$

- (a) Find the later time and determine the value of g .
- (b) Find the velocity of the object at that time.

C. Arc Length and Distance Along a Curve

Consider the curve

$$\mathbf{r}(t) = \langle 2(\cos t + t \sin t), 2(\sin t - t \cos t), t^2 \rangle, \quad t \geq 0.$$

- (a) Find the distance traveled along the curve from $t = 1$ to $t = 3$.
- (b) Starting at $\mathbf{r}(0)$ and moving in the direction of increasing t , find the point P that lies $9\sqrt{2}$ units along the curve from $\mathbf{r}(0)$.

D. Tangential and Normal Acceleration and Curvature

A particle moves along the curve

$$\mathbf{r}(t) = \left\langle t + \frac{t^3}{3}, t - \frac{t^3}{3}, t^2 \right\rangle, \quad t \geq 0.$$

- (a) Find the tangential component of acceleration $a_T(t)$.
- (b) Find the normal component of acceleration $a_N(t)$.
- (c) Find the curvature $\kappa(t)$.

Answer Key

Final answers only

A. Position, Velocity, and the Osculating Plane

(a)

$$\mathbf{r}(t) = \left\langle -2 \cos t + t + 2, t^3 + \frac{1}{2}t^2 + 2t + 1, -2t^2 + t - 1 \right\rangle$$

and

$$\mathbf{r}(1) = \left\langle 3 - 2 \cos 1, \frac{9}{2}, -2 \right\rangle.$$

(b)

$$-3x + 2y - z = 3$$

B. Motion with Constant Acceleration

(a)

$$t = 2, \quad g = -16.$$

(b)

$$\mathbf{v}(2) = \langle 6, -32, 8 \rangle$$

C. Arc Length and Distance Along a Curve

(a)

$$8\sqrt{2}$$

(b)

$$P = (2(\cos 3 + 3 \sin 3), 2(\sin 3 - 3 \cos 3), 9)$$

D. Tangential and Normal Acceleration and Curvature

(a)

$$a_T(t) = 2\sqrt{2}t$$

(b)

$$a_N(t) = 2$$

(c)

$$\kappa(t) = \frac{1}{(1 + t^2)^2}$$